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Jacobi Theta Functions, Halphen Systems, and $D(2,1;\alpha)$ Symmetry

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Abstract: We study the differential structure of Jacobi theta functions from the viewpoint of the Halphen and Ramanujan systems. After a rescaling of the modular parameter, the three theta-function sectors form a natural three-fold system whose relative coordinates determine the parameter α of the exceptional Lie superalgebra $D(2,1;\alpha)$. We interpret this structure geometrically as a nontrivial, parameter-dependent bundle over the modular direction, rather than as a trivial bundle with a constant symmetry action. The resulting three sl_2 structure provides a natural connection with the R-symmetry structure of four-dimensional $N = 4$ supersymmetric theories. We further show that a purely Cartan-valued connection only captures the linearized evolution, suggesting that the full nonlinear Halphen system requires a non-Abelian superconnection.

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Contents

1 From relation of Jacobian theta function to Halplane system **1**

2 Related with the supergroup $D(2,1;\alpha)$ **4**

3 Conclusion and Discussion **6**

1 From relation of Jacobian theta function to Halplane system

We start from the Jacobi theta relation

$$\vartheta_3^4(\tau) = \vartheta_2^4(\tau) + \vartheta_4^4(\tau) \tag{1.1}$$

Suppose that under differentiation we have

$$(B-A)X_1 = (C-B)X_3 \tag{1.2}$$

let

$$X_1 = \vartheta_2^4(\tau) \tag{1.3}$$

and set

$$A = \frac{1}{4}\frac{X_1'}{X_1} = \frac{\vartheta_2'}{\vartheta_2} \tag{1.4}$$

Taking the derivative gives

$$4\vartheta_3^3\vartheta_3' = 4\vartheta_2^3\vartheta_2' + 4\vartheta_4^3\vartheta_4' \tag{1.5}$$

which is equivalent to

$$\vartheta_3^4 B = \vartheta_2^4 A + \vartheta_4^4 C \tag{1.6}$$

Using

$$\vartheta_2(\tau) = 2q^{1/4}\prod_{n\geq 1}(1-q^{2n})(1+q^{2n})^2 \quad q = e^{i\pi\tau} \tag{1.7}$$

we have

$$\log\vartheta_2 = \log 2 + \frac{1}{4}\log q + \sum_{n\geq 1}\log(1-q^{2n}) + 2\sum_{n\geq 1}\log(1+q^{2n}) \tag{1.8}$$

the derivative is

$$\partial_\tau = i\pi q\partial_q \tag{1.9}$$

thus

$$\partial_\tau \log\vartheta_2 = \frac{i\pi}{4} + i\pi\sum_{n\geq 1}\left(\frac{-2nq^{2n}}{1-q^{2n}} + \frac{4nq^{2n}}{1+q^{2n}}\right) \tag{1.10}$$

Using the identtities

$$\sum_{n\geq 1}\frac{nq^n}{1-q^n}=\sum_{m\geq 1}\sigma_1(m)q^m \quad \sum_{n\geq 1}\frac{nq^n}{1+q^n}=\sum_{m\geq 1}(-1)^{m-1}\sigma_1(m)q^m \tag{1.11}$$

Therefore,

$$A=\partial_\tau \log\vartheta_2=\frac{i\pi}{12}(E_2(\tau)+\vartheta_3^4+\vartheta_4^4) \tag{1.12}$$

$$B=\partial_\tau \log\vartheta_3=\frac{i\pi}{12}(E_2(\tau)+\vartheta_2^4-\vartheta_4^4) \tag{1.13}$$

$$C=\partial_\tau \log\vartheta_4=\frac{i\pi}{12}(E_2(\tau)-\vartheta_2^4-\vartheta_3^4) \tag{1.14}$$

then

$$A-B=\partial_\tau \log\vartheta_2-\partial_\tau \log\vartheta_3=\frac{i\pi}{4}X_3 \tag{1.15}$$

$$B-C=\partial_\tau \log\vartheta_3-\partial_\tau \log\vartheta_4=\frac{i\pi}{4}X_1 \tag{1.16}$$

$$C-A=\partial_\tau \log\vartheta_4-\partial_\tau \log\vartheta_2=-\frac{i\pi}{4}X_2 \tag{1.17}$$

Taking the derivative, we obtain

$$A'-B'=\frac{i\pi}{4}X_3'=4C(A-B) \tag{1.18}$$

$$B'-C'=\frac{i\pi}{4}X_1'=4A(B-C) \tag{1.19}$$

$$C'-A'=-\frac{i\pi}{4}X_2'=4B(A-C) \tag{1.20}$$

At the same time,

$$X_3'=4X_3C \tag{1.21}$$

$$X_1'=4X_1A \tag{1.22}$$

$$X_2'=4X_2B \tag{1.23}$$

since the equation have a symmetric form, we first guess the first-order system can be written as

$$A'=-BC+A(B+C) \tag{1.24}$$

$$B'=-CA+B(C+A) \tag{1.25}$$

$$C'=-AB+C(A+B) \tag{1.26}$$

try check it

$$A'-B'=2C(A-B) \tag{1.27}$$

the difference is the normalization, for most easies way we normalize the derivative first rescale

$$\tilde{\tau}=2\tau \tag{1.28}$$

then

$$\partial_{\tilde{\tau}} = \frac{1}{2}\partial_\tau \tag{1.29}$$

and under S, T modular transformation, the set

$$\{\vartheta_i(2\tau), \vartheta_i(\frac{\tau}{2}), \vartheta_i(\frac{\tau+1}{2})\}_{i=2,3,4} \tag{1.30}$$

is closed. And it is $\Gamma_0(2) \setminus SL_2(\mathbb{Z})$ and

$$X_0(2) = \Gamma_0(2) \setminus \mathbb{H}^* \tag{1.31}$$

and

$$\lambda(\tau) = \frac{\vartheta_2^4(\tau)}{\vartheta_3^4(\tau)} \quad \lambda(2\tau) \quad \lambda(\frac{\tau}{2}) \quad \lambda(\frac{\tau+1}{2}) \tag{1.32}$$

is the standard Hauptmodul. the modular curve

$$X_0(2) \to X(1) \tag{1.33}$$

has degree 3.

We now change variables. First,

$$\tilde{\tau} = 2\tau \quad \partial_{\tilde{\tau}} = \frac{1}{2}\partial_\tau \tag{1.34}$$

choose

$$Y_1 = \vartheta_2^4(\tilde{\tau}) \tag{1.35}$$

and

$$\alpha = \partial_{\tilde{\tau}} \log \vartheta_2(\tilde{\tau}) = \frac{1}{2}\partial_\tau \log \vartheta_2(\tilde{\tau}) \tag{1.36}$$

$$\beta = \frac{1}{2}\partial_\tau \log \vartheta_3(\tilde{\tau}) \tag{1.37}$$

$$\gamma = \frac{1}{2}\partial_\tau \log \vartheta_4(\tilde{\tau}) \tag{1.38}$$

and

$$E_2 = \frac{1}{3}(\vartheta_2^4 + \vartheta_3^4 + \vartheta_4^4) \tag{1.39}$$

$$E_4 = \frac{1}{2}(\vartheta_2^8 + \vartheta_3^8 + \vartheta_4^8) \tag{1.40}$$

$$E_6 = \frac{1}{2}(\vartheta_2^4 + \vartheta_3^4)(\vartheta_3^4 + \vartheta_4^4)(\vartheta_4^4 + \vartheta_2^4) \tag{1.41}$$

so

$$E_2(\tilde{\tau}) = \frac{1}{3}(Y_1 + Y_2 + Y_3) \tag{1.42}$$

$$E_4(\tilde{\tau}) = \frac{1}{2}(Y_1^2 + Y_2^2 + Y_3^2) \tag{1.43}$$

$$E_6(\tilde{\tau}) = \frac{1}{2}(Y_1 + Y_2)(Y_2 + Y_3)(Y_3 + Y_1) \tag{1.44}$$

thus

$$\alpha' = -\beta\gamma + \alpha(\beta+\gamma) \tag{1.45}$$
$$\beta' = -\gamma\alpha + \beta(\gamma+\alpha) \tag{1.46}$$
$$\gamma' = -\alpha\beta + \gamma(\alpha+\beta) \tag{1.47}$$

and

$$\alpha(\tilde{\tau}) = \partial_{\tilde{\tau}} \log \vartheta_2 = \frac{1}{2}\partial_\tau \log \vartheta_2 = \frac{i\pi}{24}(E_2 + Y_2 + Y_3) \tag{1.48}$$
$$\beta(\tilde{\tau}) = \partial_{\tilde{\tau}} \log \vartheta_3 = \frac{1}{2}\partial_\tau \log \vartheta_3 = \frac{i\pi}{24}(E_2 + Y_1 - Y_3) \tag{1.49}$$
$$\gamma(\tilde{\tau}) = \partial_{\tilde{\tau}} \log \vartheta_4 = \frac{1}{2}\partial_\tau \log \vartheta_4 = \frac{i\pi}{24}(E_2 - Y_1 - Y_2) \tag{1.50}$$

In particular,

$$\alpha - \beta = \frac{i\pi}{8} Y_3 \tag{1.51}$$
$$\beta - \gamma = \frac{i\pi}{8} Y_1 \tag{1.52}$$
$$\gamma - \alpha = -\frac{i\pi}{8} Y_2 \tag{1.53}$$
$$\tag{1.54}$$

do the derivative of E_2 E_4 E_6

$$E_2(\tilde{\tau})' = \frac{1}{3}(Y_1' + Y_2' + Y_3') \tag{1.55}$$
$$E_4'(\tilde{\tau}) = \frac{1}{2}(2Y_1Y_1' + 2Y_2Y_2' + 2Y_3Y_3) \tag{1.56}$$
$$E_6'(\tilde{\tau}) = \frac{1}{2}\{(Y_1' + Y_2')(Y_2 + Y_3)(Y_3 + Y_1) + (Y_1 + Y_2)(Y_2' + Y_3')(Y_3 + Y_1) + (Y_1 + Y_2)(Y_2 + Y_3)(Y_3' + Y_1')\} \tag{1.57}$$

2 Related with the supergroup $D(2,1;\alpha)$

We first perform the following renormalization of the variables:

$$\widetilde{Y}_1 = \frac{Y_1}{\sum_i Y_i}. \tag{2.1}$$

Then

$$\widetilde{Y}_1 + \widetilde{Y}_2 + \widetilde{Y}_3 = 1 \tag{2.2}$$

We set

$$\alpha(\tau) = \frac{\widetilde{Y}_2}{\widetilde{Y}_1}, \tag{2.3}$$

and then

$$1 + \alpha = -\frac{\widetilde{X}_3}{\widetilde{X}_1}. \tag{2.4}$$

After this renormalization of the variables, we consider the supergroup $D(2,1;\alpha)$. Its bosonic part is

$$g_0 = sl_2^{(1)} \oplus sl_2^{(2)} \oplus sl_2^{(3)}. \tag{2.5}$$

We denote the generators by

$$J_+^{(a)}, J_-^{(a)}, J_0^{(a)}, \qquad a = 1,2,3. \tag{2.6}$$

We take

$$X_1 = 2\partial_\tau \log \vartheta_2. \tag{2.7}$$

We now encode the three variables into the Cartan part by

$$A^{(a)} = X_a J_0^{(a)}. \tag{2.8}$$

Thus the bosonic connection is

$$A_\tau = \sum_{a=1}^{3} X_a J_0^{(a)}. \tag{2.9}$$

We now introduce the Clifford generators

$$\{\gamma_i, \gamma_j\} = 2\delta_{ij}. \tag{2.10}$$

We define

$$Q_1 = \sqrt{\tilde{X}_1}, \gamma_1 \otimes E_1, \quad Q_2 = \sqrt{\tilde{X}_2}, \gamma_2 \otimes E_2, \quad Q_3 = \sqrt{\tilde{X}_3}, \gamma_3 \otimes E_3. \tag{2.11}$$

Here E_i are the intertwiners between the three $sl_2^{(a)}$ factors. We impose

$$\{E_i, E_j\} = \delta_{ij} J_0^{(i)}. \tag{2.12}$$

Then

$$\{Q_i, Q_i\} = 2X_i J_0^{(i)}. \tag{2.13}$$

Therefore,

$$Q = \sum_{i=1}^{3} Q_i, \qquad Q, Q = 2\sum_{i=1}^{3} X_i J_0^{(i)}. \tag{2.14}$$

We now ask whether this construction is consistent with the evolution of the algebraic structure. Namely, we consider

$$\partial_\tau Q + [A, Q] = 0; ? \tag{2.15}$$

We first check the derivative of each Q_i:

$$\partial_\tau Q_i = \frac{1}{2}\frac{X_i'}{X_i} Q_i. \tag{2.16}$$

On the other hand,

$$[A, Q_i] = \sum_j \widetilde{X} * j[J_0^{(j)}, Q_i] = \left(\sum_j \alpha * ij X_j\right) Q_i, \tag{2.17}$$

where the coefficients depend on the representation. Therefore, the evolution equation becomes

$$\frac{1}{2}\frac{X_i'}{X_i} + \sum_j \alpha_{ij} X_j = 0. \tag{2.18}$$

Equivalently,

$$\frac{X_i'}{X_i} = -2\sum_j \alpha_{ij} X_j. \tag{2.19}$$

This equation is linearized in the variables X_i, while the Halphen system is quadratic. Therefore, the above construction does not directly reproduce the Halphen system, and we should try another construction.

There is also a possible relation with the degeneration of $D(2,1;\alpha)$. At $\alpha \mapsto 0$,

$$D(2,1;\alpha) \to \mathfrak{osp}(4|2). \tag{2.20}$$

Further,

$$\mathfrak{osp}(4|2) \to \mathfrak{osp}(2|1) \oplus \mathfrak{osp}(2|1) \to A_2. \tag{2.21}$$

Finally, we consider the behavior in characteristic 2. $D(2,1;\alpha)$ becomes a family of Lie algebras if the characteristic of the field is 2. The group has three exceptional orbits

$$0, -1, \infty \qquad 1, -2, -\frac{1}{2} \qquad -\frac{1}{2}, \pm\frac{\sqrt{3}}{2}. \tag{2.22}$$

Then $D(2,1)$ is the second orbit. The super algebra is the third orbit and admits an outer automorphism of order 3.

3 Conclusion and Discussion

The theta-function realization provides a natural differential system for the three quantities

$$Y_1 = \vartheta_2^4, \qquad Y_2 = \vartheta_3^4, \qquad Y_3 = \vartheta_4^4.$$

After the rescaling

$$\tilde{\tau} = 2\tau,$$

their logarithmic derivatives satisfy the Halphen system, which is equivalent to the Ramanujan differential system after expressing the symmetric combinations in terms of E_2, E_4, E_6. The three-fold structure of the theta functions then suggests a natural relation with the three sl_2 factors appearing in

$$D(2,1;\alpha)_{\bar{0}} \simeq sl_2^{(1)} \oplus sl_2^{(2)} \oplus sl_2^{(3)}.$$

The important point is that the parameter α is already encoded in the relative theta-function coordinates. In particular,

$$\tilde{Y}_1 + \tilde{Y}_2 + \tilde{Y}_3 = 1, \qquad \alpha(\tau) = \frac{\tilde{Y}_2}{\tilde{Y}_1},$$

so that the $D(2,1;\alpha)$ parameter is not introduced as an additional external constant. It is determined by the modular coordinate and by the relative position of the three theta-function sectors.

This also gives a natural interpretation in the context of four-dimensional $N = 4$ supersymmetry. The $N = 4$ theory has a nontrivial R-symmetry structure, and its super-symmetry algebra naturally involves the action of the R-symmetry on the supercharges. From this point of view, the appearance of the three sl_2 factors and their parameter-dependent organization in $D(2,1;\alpha)$ is not an accidental algebraic feature of the Halphen system. Rather, it provides a natural super-symmetric extension of the modular differential system. In particular, the theta-function coordinates can be viewed as coordinates on a nontrivial bundle carrying the corresponding $D(2,1;\alpha)$ R-symmetry structure.

The bundle obtained in this way is not a trivial bundle with a constant $D(2,1;\alpha)$ structure. The theta functions depend on the modular coordinate $\tilde{\tau}$, and consequently the corresponding parameters and local frames vary along the modular direction. The Halphen equations describe precisely this variation. Thus the natural geometric object is a nontrivial, parameter-dependent $D(2,1;\alpha)$-bundle rather than a product bundle equipped with a constant algebraic action.

We also found that a purely Cartan-valued connection is not sufficient to reproduce the full nonlinear Halphen system. The corresponding parallel-transport equation only gives the linearized form of the evolution. This indicates that the nonlinear information should be carried by the off-diagonal and, more generally, non-Abelian components of the full superconnection. The explicit construction of this full $D(2,1;\alpha)$-valued superconnection remains an interesting next step.

Therefore, the main structure obtained here can be summarized as

$$\text{Jacobi theta functions} \longrightarrow (Y_1, Y_2, Y_3) \longrightarrow \text{Halphen system} \longrightarrow D(2,1;\alpha)\text{-bundle}.$$

The theta functions provide the modular coordinates, the Halphen equations describe their evolution, and the parameter α organizes the corresponding $D(2,1;\alpha)$ R-symmetry structure. The resulting bundle is nontrivial precisely because these data vary over the modular direction. This gives a geometric way to connect the theta-function differential system with the parameter-dependent super-symmetric structure relevant to $N = 4$ theories.